

F-24-170-D Kisan Rabta

Project Team

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Chapter 1

Introduction

Agriculture is a cornerstone of Pakistan's economy. Farmers play an essential role in the nation's development and stability. However, these farmers need assistance with various issues, including water availability, agricultural diseases, necessary agricultural information, and access to the market. Unfortunately, no existing system can collect these problems in real-time and analyze these concerns so that necessary actions can be taken, depending on the language, such as Urdu. Sybrid Islamabad found this problem and needed a system that answered these farmers' concerns and ensured their voices were heard. Necessary actions are taken against them, creating something constructive for the farmers.

1.1 Problem Statement

This project comprehensively collects, processes, and analyses data in real time in response to the needs of the Urdu-speaking farming population within Pakistan. A mobile application enables workers in the field to record conversations with farmers that can later be transcribed and analyzed using machine learning models built in Python. It will have an online database comprising agricultural information, and problems which are visually represented through our web interface. The other facilities included in this system are an admin panel for running extensive data queries and filtering for displaying relevant information thereby making decisions based on logical reasoning, hence improving farmers' production ability and making them more productive.

1.2 Scope

This project comprehensively collects, processes, and analyses data in real time in response to the needs of the Urdu-speaking farming population within Pakistan. A mobile application enables workers in the field to record conversations with farmers that can later be transcribed and analyzed using machine learning models built in Python. It will have an online database comprising agricultural information, and problems which are visually represented through our web interface. The other facilities included in this system are an admin panel for running extensive data queries and filtering for displaying relevant information thereby making decisions based on logical reasoning, hence improving farmers' production ability and making them more productive.

1.3 Modules

Our project, Kisan Rabta, aims to transform agriculture data collection and analysis to use advanced modern-day technologies by integrating with field personnel and admin management. There will be two main modules of our project. One is the mobile app, and the other is the dynamic Web interface. Our mobile has simple functionalities that will streamline the data collection process to collect the data accurately and in real-time. This data is then stored in our database for further operations. Here comes our machine learning models into action. They will use this data to transcribe the extensive recordings into text, and then by using the agriculture data, we will remove the noise from the data and keep only agriculture-related things. The goal of each Module of our project is to make efficiency improvements and simplify the workflow. Following is a detailed description of our modules with relevant features.

1.3.1 Client Mobile App

Field agents can install an application on their phones that gives them a simple interface for recording conversations in Urdu. This tool gathers data about agriculture within a timeline.

1. **Secure Login:** To make the system more secure, there will be an authentication mechanism for secure login. To this end, the client cannot sign up directly through the app. Only the admins can create login credentials for the field personnel, who can then log in using these credentials.
2. **Profile Management:** Users can access their profiles' information and carry out profile management tasks, such as changing their profile pictures.

3. **Voice Recording:** High-quality voice recordings in real-time are used to store comprehensive discussions involving field agents and farmers
4. **Upload Integration:** After the client pushes these recordings into the database. These extensive conversation recordings are then stored in our database for further operations.
5. **Audio File Management:** These recordings are managed on the mobile until the field personnel does not upload them to the database.

1.3.2 Admin Web App

For the second module of our project “Kisan Rabta,” we are building a dynamic web interface for admins to visualize, query, and filter based on the requirements.

1. **Secure Login:** Admin Login Secure admin login functionality will ensure that no unauthorized users can access the dashboard.
2. **Admin Panel:** A secure Admin Panel for managing user activity, creating login credentials for field employees, and managing their activities.
3. **Data Visualization Tools:** Interactive tools for visualizing agricultural data with graphs, charts, and maps will be provided. Users can see the agriculture data specific to a particular area. The admin can also search, filter, and query data. For example, suppose the admin wants to see the agricultural information in the Punjab province. In that case, the admin can write Punjab in the search bar, and our dashboard will show them the map of Punjab province, highlighting the cities of Punjab and pinpointing the cities where the admin can select any city for more information.

1.4 User Classes and Characteristics

Identify the various user classes that you anticipate will use this product, and describe their pertinent characteristics.

User class	Description
Field Agent (App User)	The primary user of the mobile application will be the Field Agent. They will collect the data from the farmers in form of audio recordings of their conversation or manually entering the details of the issue and keep the location monitored by geo tagging for easy labelling of data.
Admin (Administer)	The admin is responsible for managing users and the system through the web portal. The admin creates and manages the field agents accounts, review the collected data and utilise the web application interface for data visualization, filtering and querying.
Farmer (Farmer Participant)	The farmer shares the data of the agricultural issue faced during the conversation with the field agent. They do not interact with the application directly, but they provide the relevant details that we need to process the issue they are facing.

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the project.

2.1 Use-case

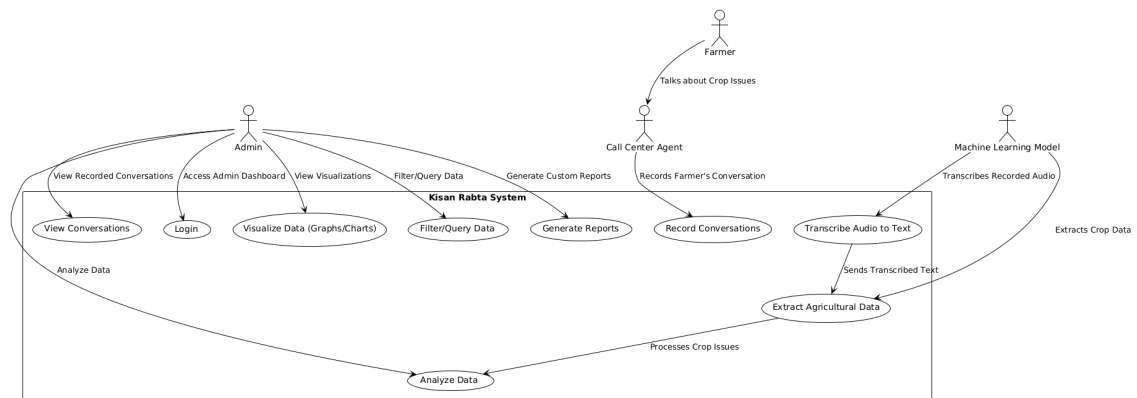


Figure 2.1: Use Case Diagram

Detailed Use cases:

ID	UC001
Name	Record Conversations
Actors	agent, farmer
Description	This use case involves recording the conversations between a farmer and the agent regarding crop issues.
Preconditions	• The agent is logged in. • The farmer is ready to engage in a conversation.
Postconditions	• The audio recording is saved in the system. • A reference to the recording is stored in the database.
Main Success Scenario	1. The agent selects the "Record Conversation" option. 2. The system starts the audio recording. 3. The agent engages with the farmer about their crop issues. 4. The system records the conversation in real-time. 5. The agent stops the recording when the conversation ends. 6. The system saves the audio file in the database.
Extension	If recording fails: The system alerts the agent to retry.

Table 2.1: Use Case: Recording Conversations

ID	UC002
Name	Transcribe Audio to Text
Actors	Machine Learning Model
Description	This use case involves transcribing recorded audio from a conversation into text using a machine learning model.
Preconditions	• Audio recording exists. • The system is capable of processing the audio file.
Postconditions	• The text transcription is saved in the system.
Main Success Scenario	system sends the recorded audio to the transcription service.The machine learning model processes the audio. The model converts the audio to text.The transcription is saved in the database.
Extension	If transcription fails: An error log is generated, and the process is retried.

Table 2.2: Use Case: Transcribe Audio to Text Use Case

ID	UC003
Name	Extract Agricultural Data
Actors	Machine Learning Model
Description	This use case involves analyzing transcribed text to extract relevant agricultural data, such as crop types, diseases, or issues mentioned by the farmer.
Preconditions	1. Transcribed text exists.
Postconditions	• Agricultural data is saved and associated with the transcription.
Main Success Scenario	1. The system sends the transcribed text to the data extraction service. 2. The machine learning model processes the text to identify key agricultural data. 3. Extracted data is stored in the database.
Extension	If data extraction fails: The system retries or flags the issue.

Table 2.3: Use Case: Extract Agricultural Data Use Case

ID	UC004
Name	Analyze Data
Actors	Admin
Description	The admin can analyze the agricultural data extracted from conversations to derive useful insights for reporting or decision-making.
Preconditions	• Agricultural data exists in the system.
Postconditions	• Insights and analysis are presented to the admin.
Main Success Scenario	1. Admin logs into the system. 2. Admin selects the "Analyze Data" option. 3. The system saves the data. 4. Admin can view the data trends and insights.
Extension	If login fails, the system displays an error message.

Table 2.4: Use Case: Analyze Data Use Case

ID	UC005
Name	Login
Actors	Admin
Description	The admin must log into the system to access any data or features.
Preconditions	• The admin has valid login credentials.
Postconditions	• The admin is logged in and has access to the system.
Main Success Scenario	1. The admin enters the username and password. 2. The system validates the credentials. 3. If valid, the admin is granted access to the dashboard.
Extension	If data collection fails: The system alerts the agent.

Table 2.5: Use Case: Login Use Case

2.2 Functional Requirements

This section outlines the functional requirements of the Kisan Rabta Project. These requirements are critical for the system to perform properly and to record the interaction between the user and the system. The requirements ensure accurate data gathering, analysis, and report generation functionalities.

2.2.1 Voice Recording Interface

The following are the requirements for **the Voice Recording Interface** module :

1. **FR 1.1:** The system must include a voice recording interface for the call center agents to capture the conversations between the farmers and them in Urdu language.
2. **FR 1.2:** The system should enable the agents to manually start and stop the recording during the conversation to ensure that the unwanted conversation is not captured.

3. **FR 1.3:** The recorded files should be kept securely in the system.

2.2.2 Audio-to-Text Conversion

The following are the requirements for **the Audio-to-Text Conversion** module:

1. **FR 2.1:** The system will use machine learning models to automatically transcribe the recorded Urdu audio conversations into text.
2. **FR 2.2:** The transcription accuracy will be high to ensure minimum data loss.

2.2.3 Agricultural Data Extraction

The following are the requirements for **the Agricultural Data Extraction** module:

1. **FR 3.1:** The system will automatically extract crucial agricultural information from the transcribed text, such as crop diseases, problems, and treatment approaches.
2. **FR 3.2:** The system must classify extracted data by crop type, disease name, and farmer area.
3. **FR 3.3:** Extracted data must be retained in a database for future analysis and reporting.

2.2.4 Data Analysis and Insights

The following are the requirements for **the Data Analysis and Insights** module:

1. **FR 4.1:** The system will evaluate the retrieved data to discover patterns, such as crop diseases, farmers' problems, and other agriculture-related information.
2. **FR 4.2:** The system will provide insights depending on area, demographics, and crop kind.
3. **FR 4.3:** The analysis must accommodate large-scale datasets for future scalability.

2.2.5 Data Visualization

The following are the requirements for **Data Visualization** module:

1. **FR 5.1:** The system must include a web-based dashboard with data visualization features such as graphs and charts.

2. **FR 5.2:** Users may pick and filter data according to area, crop type, disease name, and time range.
3. **FR 5.3:** The system must show visual representations of data trends, such as bar charts, to aid decision-making.

2.2.6 Admin Dashboard and Reporting

The following are the requirements for **Admin Dashboard and Reporting** module:

1. **FR 6.1:** The system must have an admin interface that allows the administrators to query data using various filters such as the region, crop type, and farmer demographic.
2. **FR 6.2:** Administrators must be able to create reports based on the filtered data.
3. **FR 6.3:** The system must enable reports to be exported in formats such as PDF or CSV for external usage.

2.2.7 User Authentication and Roles

The following are the requirements for **User Authentication and Roles** module:

1. **FR 7.1:** The system must offer a login interface for all users, including agents and administrators.
2. **FR 7.2:** The system must allow role-based access control, with administrators having full access and contact agents having restricted access depending on their function.
3. **FR 7.3:** Users will be authorized using their login and password.

2.3 Non-Functional Requirements

Following are the Non-Functional Requirements of Kisan Rabta Project.

2.3.1 Performance

1. **PER 1.1:** The system must support the concurrent use of up to 15 field agents and administrators without significant degradation in performance.

2. **PER 1.2:** Audio-to-text transcription should be processed within 5 minutes of uploading, even with large audio files.
3. **PER 1.3:** The system must handle the storage and processing of large datasets, ensuring that data retrieval queries for visualization take time between 30 - 60 seconds.

2.3.2 Scalability

1. **SCA 2.1:** The system must be scalable to accommodate future growth, supporting up to 25 field agents and an increase in data collection without significant changes in performance.

2.3.3 Security

1. **SEC 3.1:** The system must implement role-based access control (RBAC), with strict authorization checks ensuring only authorized users (field agents and administrators) have access to the data.
2. **SEC 3.2:** User authentication must be strong and include multi-factor authentication for administrators.

2.3.4 Usability

1. **USA 4.1:** The mobile and web interfaces must be user-friendly and intuitive, allowing field agents and administrators to navigate through the system with minimal training.

2.3.5 Data Integrity

1. **DAI 5.1:** The system must ensure that all data, including audio recordings, transcriptions, and reports, is stored and maintained in its correct form.

2.3.6 Maintainability

1. **MAI 6.1:** The system architecture must allow for easy updates and maintenance, ensuring that new features and security patches can be deployed with minimal downtime.

2. **MAI 6.2:** Codebase should follow modular architecture patterns to allow individual components, like machine learning models or the dashboard, to be updated independently.

2.3.7 Compliance

1. **COM 7.1:** All audio recordings must adhere to consent-based data collection practices, with appropriate notifications provided to farmers before recording.

2.4 Domain Model

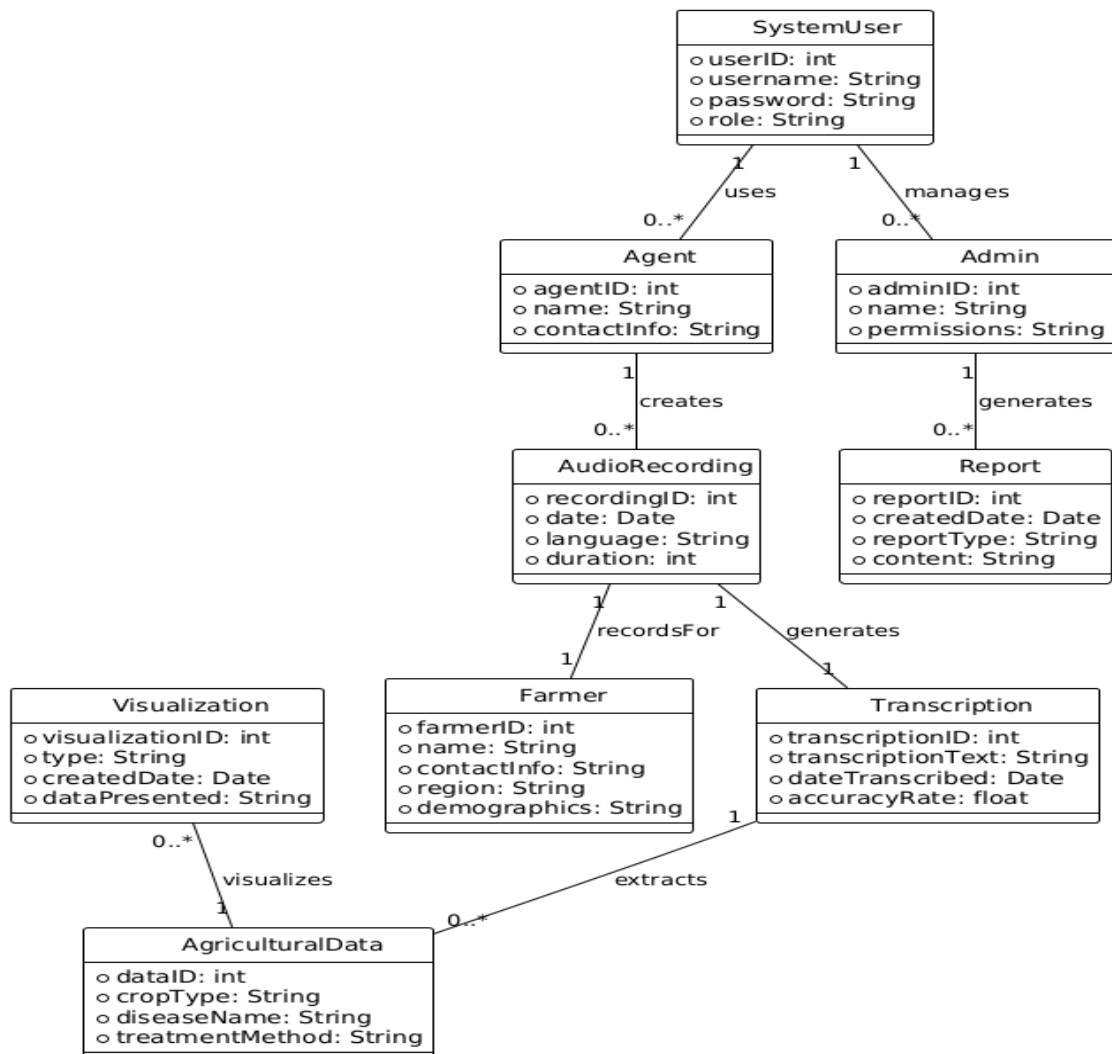


Figure 2.2: Domain Model

Chapter 3

System Overview

3.1 Architectural Design

1. Presentation Layer (User Interface)

The Presentation Layer manages user interactions across both mobile and web platforms. It serves as the interface for agents and administrative users to interact with the system, facilitating seamless communication and user engagement.

2. Application Layer (Business Logic)

The Application Layer handles the core business logic of the system, including voice recording, transcription processing, and user management. It is also responsible for invoking machine learning models to transcribe audio inputs and extract relevant agricultural information.

3. Data Layer

The Data Layer ensures that all collected and processed data is securely stored and easily accessible for future use, maintaining data integrity and availability across the system.

4. Machine Learning Layer

The Machine Learning Layer contains models for audio-to-text conversion and agricultural data extraction. It interacts with the Application Layer to provide transcription-based analysis and insights that aid in decision-making and information processing.

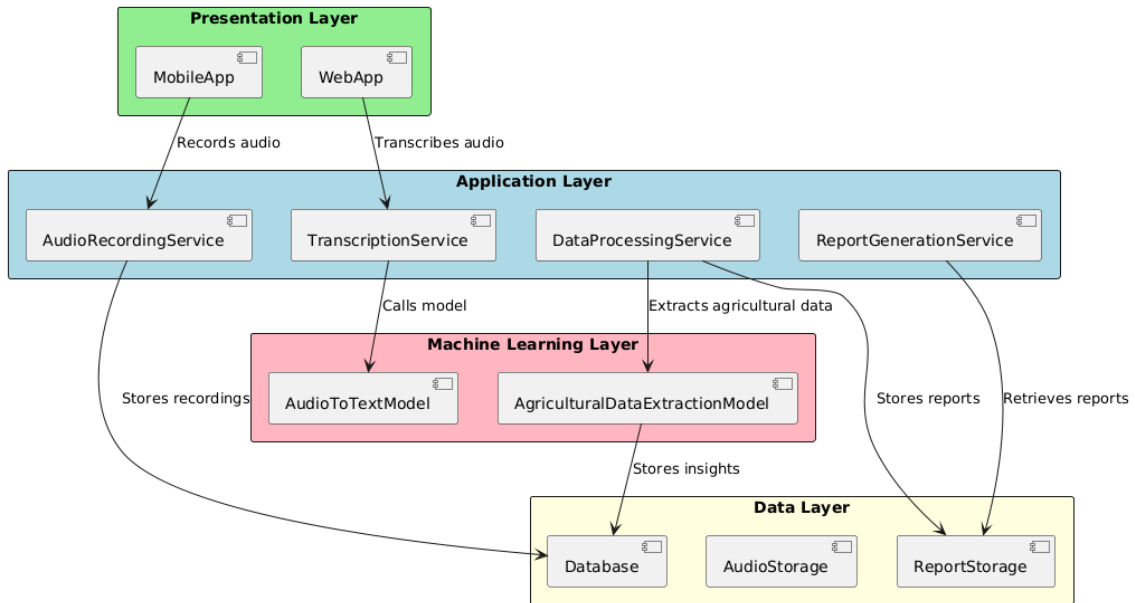


Figure 3.1: Architecture Diagram

3.2 Design Models

Design Models for Object-Oriented Development Approach

- Activity Diagram

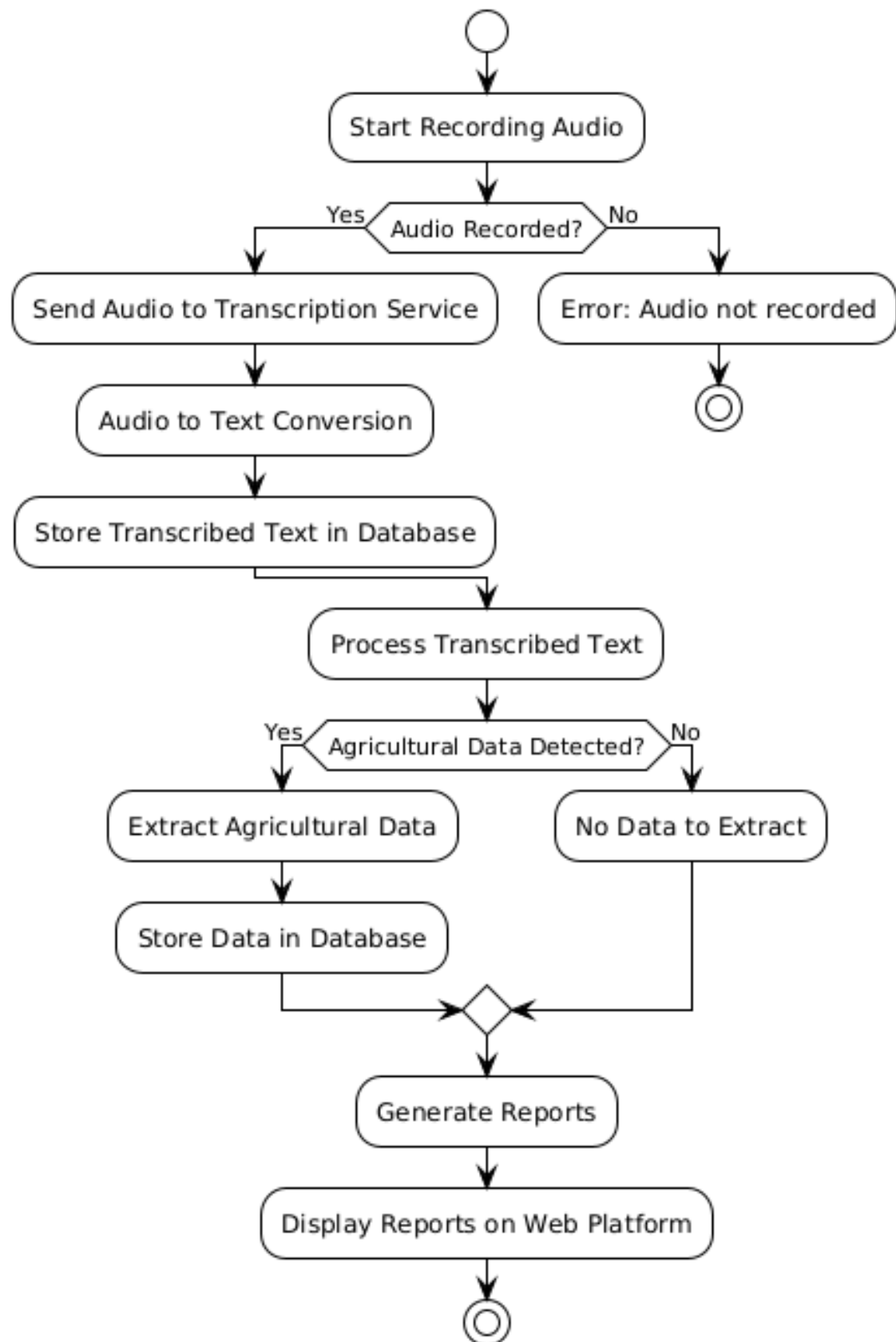


Figure 3.2: Activity Diagram

• Class Diagram

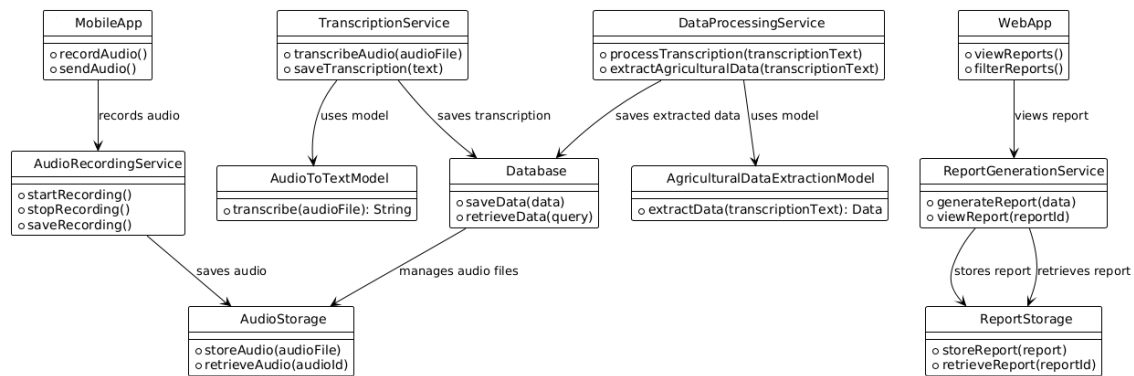


Figure 3.3: Class Diagram

• Class-level Sequence Diagram

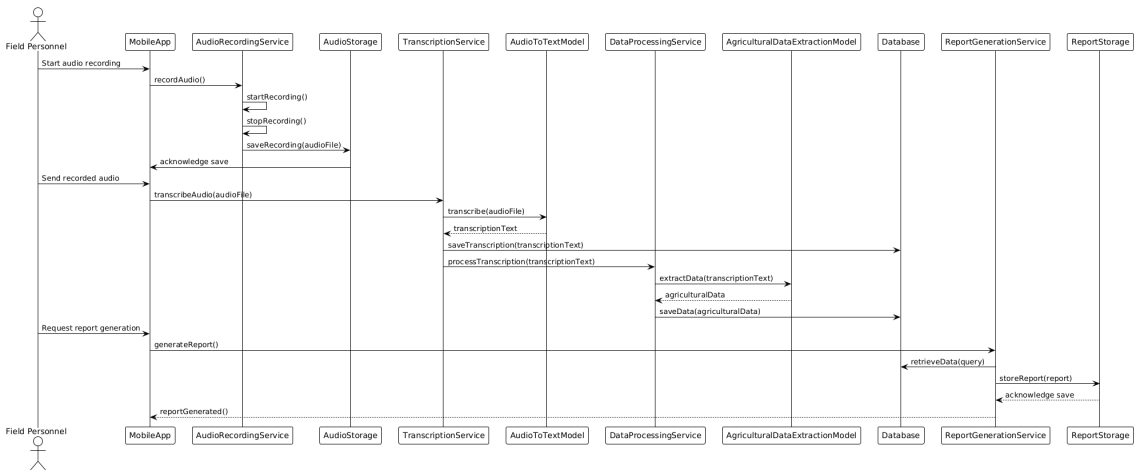


Figure 3.4: Class Level Sequence Diagram

• State Transition Diagram

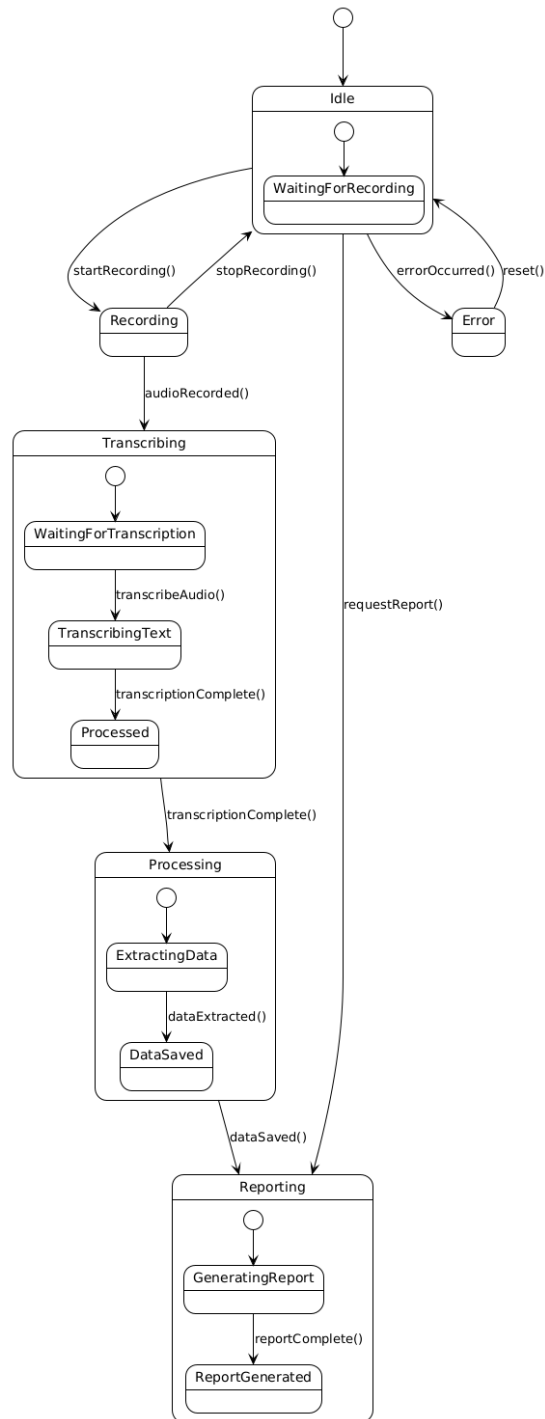


Figure 3.5: State Transition Diagram

3.3 Data Design

The information domain is transformed into the data structures by organizing all the entities into the form of collections and documents in the non-relational database MongoDB.

Each collection in the database represents a specific entity and each document represents all the attributes of that entity. This allows easy access, retrieval, and manipulation of the data.

- **Entity Relationship Diagram**

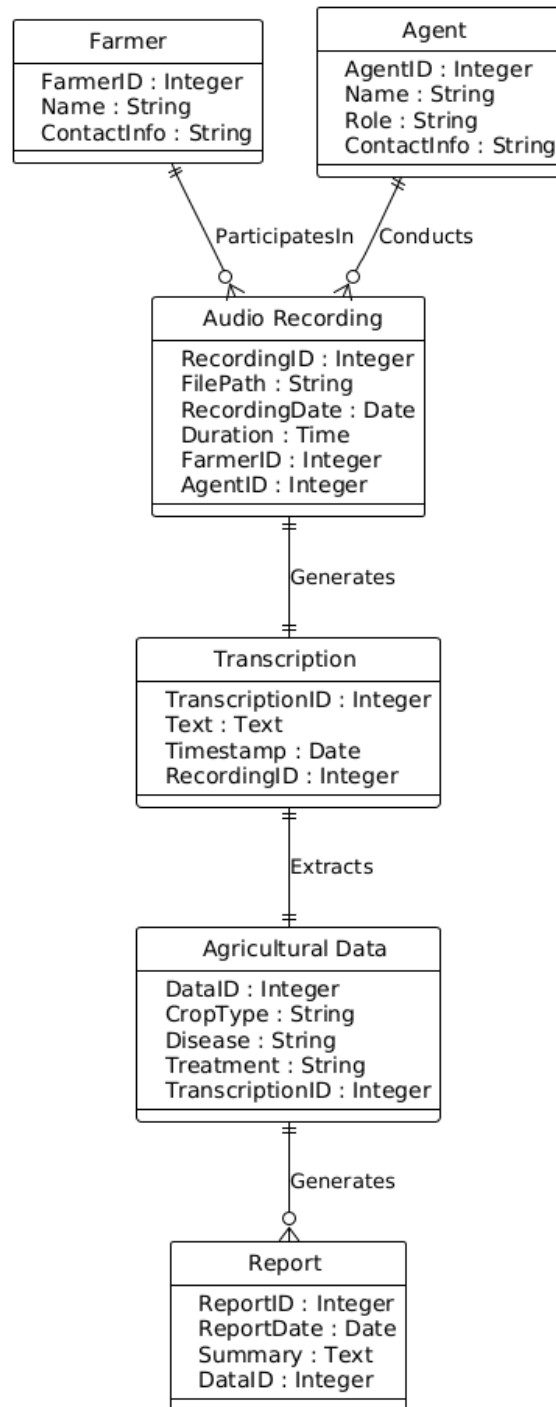


Figure 3.6: Entity Relationship Diagram

Storage, Processing, and Organization of Major Entities

1. Farmers Collection:

- **Storage:** Each Farmers information is stored in the farmers document having attribute Unique identifier, Name etc.
- **Processing:** When a farmer interacts with the system i.e records audio conversation, their information can be easily accessible by the system.
- **Organization:** All farmer documents are stored in the Farmers collection.

2. Agents Collection:

- **Storage:** Each agents information is stored in the agents document having attribute Unique identifier, Name, Role, Contact Information etc.
- **Processing:** When an agent interacts with the system i.e records audio conversation or pushes the recording onto the database, their information can be easily accessible by the system.
- **Organization:** All agent documents are stored in the agents collection.

3. Audio Recordings Collection:

- **Storage:** Each Audio Recordings information is stored in the Audio Recordings document having attributes like RecordingID, FilePath, RecordingDate, Duration, and references to FarmerID and AgentID.
- **Processing:** When a recording is created, it generates a new document with the name Audio Recording linking it to the respective farmer and agent. This allows for efficient retrieval of all related recordings.
- **Organization:** All audio recording documents are stored in the Audio Recordings collection.

4. Transcriptions Collection:

- **Storage:** Each Transcriptions information is stored in the Transcriptions document having attributes like TranscriptionID, Text, Timestamp, and a reference to RecordingID.
- **Processing:** Each transcription can be generated from its corresponding audio recording, helping the system maintain a clear link for processing and retrieval.
- **Organization:** All transcription documents are stored in the Transcriptions collection.

5. Agricultural Data Collection:

- **Storage:** This collection contains documents with DataID, Crop Type, Disease, Treatment, and a reference to TranscriptionID.
- **Processing:** Data is extracted from transcriptions to create records of agricultural information, facilitating analysis and reporting.
- **Organization:** All agricultural data entries are maintained in the Agricultural Data collection.

6. Reports Collection:

- **Storage:** Each Reports information is stored in the Reports document having attributes like ReportID, ReportDate, Summary, and a reference to DataID.
- **Processing:** Reports can be generated based on the agricultural data, summarizing the findings for farmers or agents.
- **Organization:** All report documents are stored in the Reports collection.

Database

- **Database:** All of these collections exist within a single MongoDB database, which acts as the central repository for all data.
- **Collections:** The specific collections include:
 - Farmers
 - Agents
 - Audio Recordings
 - Transcriptions
 - Agricultural Data
 - Reports

This organization allows for efficient querying and data management, making it easy to handle complex relationships and large volumes of data within the system.

Bibliography